

# Marion Fumani Maphali

Network Engineer · Backend Developer · IT Professional

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## PROFESSIONAL SUMMARY

IT Communication Networks graduate from Nelson Mandela University with hands-on experience in enterprise network design, Cisco routing and switching, and Python backend development. Completed a 12-month backend developer internship at DataTank BI Solutions, building data-driven systems, tracking dashboards, and a ticketing web application within a professional BI environment. Delivered iterative development cycles responding to feedback from supervisors, management, and clients to continuously improve system functionality. Holder of multiple Cisco Networking Academy certifications covering networking, cybersecurity, IoT, and enterprise automation. Driven by a passion for continuous learning, technical problem-solving, and building reliable infrastructure.

## PROFESSIONAL EXPERIENCE

### Python Backend Developer Intern

Jul 2024 – Jul 2026

DataTank BI Solutions · North Riding, Randburg

Hosted by MICTSETA with Lumphindi Business Solutions | 12-Month Internship

- Developed backend software systems using Python for business intelligence applications in a live enterprise environment.
- Built and maintained database interactions and backend business logic for BI data processing pipelines.
- Designed and developed a web application featuring a ticketing system and BI tracking dashboards, iterating functionality based on feedback from supervisors, management, and clients.
- Provided system support and implemented technical solutions across multiple BI projects.
- Collaborated with cross-functional technical teams on software development tasks, debugging, and optimization.
- Applied analytical thinking to design reliable, efficient backend systems aligned with client requirements.

### Network Design Engineer (Academic Project)

2023 – 2024

Nelson Mandela University · Port Elizabeth, South Africa

IT Communication Networks Capstone Project | 12-Month Academic Project

- Designed and implemented a full-scale enterprise airport network infrastructure as part of the final-year capstone project.
- Applied Cisco routing and switching technologies (OSPF, VLANs, BGP, ACLs) across a complex multi-zone topology.
- Implemented VPN, IPSec, NAT/PAT, firewall rules, and network segmentation for security and traffic optimization.
- Produced comprehensive technical documentation, network diagrams, and architecture specifications.
- Simulated and validated the network design using Cisco Packet Tracer and virtual machine environments.

## EDUCATION

### Diploma in Information Technology: Communication Networks

2021 – 2025

Nelson Mandela University (NMU) · Port Elizabeth

Graduated: April 2025 · Specialization: IT Communication Networks & Enterprise Infrastructure

### National Senior Certificate (NSC / Matric)

2020

E.P.P. Mhinga High School · Limpopo

Key subjects: English, Mathematics, Life Sciences, Physical Sciences, Accounting, Xitsonga

## CERTIFICATIONS

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|---------------------------------------------------------------|------------------------------|----------|
| <b>CCNA: Enterprise Networking, Security &amp; Automation</b> | Cisco NetAcad / NMU          | Jan 2025 |
| <b>CCNAv7: Introduction to Networks</b>                       | Cisco Networking Academy     | Dec 2021 |
| <b>Introduction to Cybersecurity</b>                          | Cisco Networking Academy     | Aug 2021 |
| <b>Introduction to IoT</b>                                    | Cisco Networking Academy     | Jul 2022 |
| <b>Diploma in IT: Communication Networks</b>                  | Nelson Mandela University    | Apr 2025 |
| <b>National Senior Certificate</b>                            | Dept. of Basic Education, SA | Nov 2020 |

## TECHNICAL SKILLS

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**Networking Protocols:** TCP/IP, OSI model, IPv4/IPv6, Subnetting, LAN, WAN, VLAN, WLAN

**Routing & Switching:** Static Routing, OSPF, BGP, EIGRP, VLANs, STP, Cisco IOS

**Network Security:** Cisco ASA Firewall, VPN (IPSec, Site-to-Site), ACLs, IDS/IPS, NAT/PAT

**Tools & Platforms:** Cisco Packet Tracer, PuTTY, VirtualBox, Azure (Networking basics)

**Programming:** Python (backend scripting & automation), C#, SQL

**Productivity:** Microsoft Office 365 (Word, Excel, PowerPoint, Outlook, Teams)

**Troubleshooting:** SSH/Telnet, Ping, Traceroute, Wireshark

**Operating Systems:** Windows (advanced), Linux (CLI basics)

## KEY PROJECTS

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### BI Web Application with Ticketing System & Tracking Dashboards · DataTank BI Solutions

Developed a full-stack web application within a professional BI environment featuring a ticketing system and real-time tracking dashboards. Iterated on features continuously based on feedback from supervisors, management, and end clients, delivering a production-ready solution that supported business intelligence operations.

### Airport Network Infrastructure Design & Implementation

Designed a complete enterprise-grade airport network as a capstone project, covering full topology planning, VLAN segmentation, routing protocol configuration (OSPF, BGP), VPN setup, and security hardening across a multi-zone environment using Cisco IOS and Packet Tracer.

### Personal Portfolio Website

Designed and built a fully responsive personal portfolio website using HTML5, CSS3, and vanilla JavaScript, featuring animated network node canvas, scroll-triggered transitions, and a contact form. Deployed on GitHub Pages.

## ADDITIONAL INFORMATION

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**Languages:** Xitsonga (Native), English (Fluent)

**Driver's Licence:** Code 10 / Class C1 (Valid until 2030)

**Portfolio:** <https://marion-mars.github.io/marion-maphali-portfolio/>

**Availability:** Open to opportunities — immediate start